NOTATION · CHEAT SHEET

Set Notation at a Glance

The marks from the notation cards across the set-theory pages: how each is read and what it means. Every group links to the full card on its page.

BASICS

$\{1, 2, 3\}$	**THE SET OF ONE, TWO, THREE** Roster form: order and repeats carry no meaning.
$\{x \mid x > 0\}$	**ALL X SUCH THAT X IS POSITIVE** Set-builder form: a variable, the bar, a condition.
$\in,\ \notin$	**IS / IS NOT AN ELEMENT OF** Membership, and its denial with a slash.
$\varnothing$	**THE EMPTY SET** The set with no members; also written $\{\}$.

SUBSETS

$\subseteq,\ \subset$	**SUBSET; PROPER SUBSET** The underbar allows $A = B$; without it, $A \neq B$.
$\supseteq,\ \supset$	**SUPERSET** The same facts read from the larger set.
$\mathcal{P}(A),\ 2^A$	**THE POWER SET OF A** The set of all subsets of A.
$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$	**THE NUMBER SYSTEMS** Naturals, integers, rationals, reals.

SET OPERATIONS

$\cup,\ \cap$	**UNION; INTERSECTION** In either set; in both sets.
$A^c,\ A',\ \overline{A}$	**THE COMPLEMENT OF A** Everything in the universal set U outside A.
$A \setminus B$	**A MINUS B** In A but not in B; also written $A - B$.
$A \triangle B$	**SYMMETRIC DIFFERENCE** In exactly one of the two sets.

CARDINALITY

$\lvert A \rvert$	**THE CARDINALITY OF A** The number of members: $\lvert\{1, 2, 3\}\rvert = 3$.
$\aleph_0$	**ALEPH-NULL** The size of $\mathbb{N}$ and every countably infinite set.
$\mathfrak{c} = 2^{\aleph_0}$	**THE CONTINUUM** The size of $\mathbb{R}$, larger than $\aleph_0$.

SAME MARKS, DIFFERENT JOB

$\lvert A \rvert$	A count for a set; an absolute value for a number; a determinant for a matrix.
$\subset$	Proper subset here; some books use it for any subset.
2^A	The power set when A is a set, not a number raised to a power.
$\{x\}$	A one-element set here; the fractional part of x in arithmetic.